One Predictor for Four Rail Faults from Vibration and Sound Alone

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***Abstract*—Cameras and track-recording cars are the usual tools for finding rail defects, but sound and vibration can be captured with cheap sensors on any train. We build a single fault predictor for four rail-fault tasks using only microphone and accelerometer data from three public datasets: rail-surface faults (superelevation and wheel burns), rail corrugation heard by an in-cabin microphone, corrugation felt at the axle box, and rail joints and switches. For each task we compare a random forest on spectral features with a fine-tuned Audio Spectrogram Transformer (AST), and the predictor uses whichever is better. Clips are cut on a distance grid rather than in time, so a fault produces the same pattern at any train speed. Over repeated training runs the predictor reaches a test macro-F1 of 0.99 on rail-surface faults, 0.92 for corrugation from the axle box and 0.89 from the microphone, both measured at a train speed not seen in training, and 0.81 on joints and switches. Labels for the joint task come from an unsupervised detector that places all nine surveyed joints and switches of the Rail-VIVID dataset inside repeatable stretches of high vibration. We also report a negative result: a single shared AST with one output head per fault was unstable across four training runs, and no run matched the separate models on more than one task.**

***Index Terms*—railway track monitoring, fault detection, vibration, acoustic analysis, audio spectrogram transformer, rail corrugation, multi-task learning**

I. Introduction

Rail defects such as corrugation, wheel burns, worn joints and poorly set superelevation raise noise, wear and maintenance cost, and some lead to derailment if they go unnoticed. Most inspection relies on dedicated measurement trains or on cameras, which are costly to run often. Microphones and accelerometers are cheap and can ride on ordinary service trains, so the signal they record could flag faults between scheduled inspections.

Public labelled data for this is scarce and fragmented. Each available dataset covers one line, one vehicle and one or two kinds of fault, recorded with different sensors at different sampling rates. This paper asks how far one can get with only that public data and only vibration and sound, with no images.

Our contributions are:

- One predictor that scores four rail-fault tasks from vibration and sound, assembled from the best model per task, with results averaged over repeated training runs on grouped splits that keep every recording (or run) in a single split.
- Distance-normalised clips: each recording is resampled onto a distance grid at a fixed reference speed, so a periodic fault such as corrugation keeps the same frequency at any train speed. With this, corrugation is detected on a run at a speed not seen in training.
- An unsupervised way to label joints and switches in Rail-VIVID, whose vibration recordings carry no labels: repeatable stretches of high vibration contain all nine surveyed joints and switches.
- A negative result: a shared multi-task AST was unstable across runs, and no run matched the separate models on more than one task.

Code to rebuild every result is public (Section VIII).

II. Related Work

Acoustic track-fault detection. Shafique et al. [1] recorded track sound with microphones on a cart on Pakistan Railways and classified normal track, superelevation and wheel burns with MFCC features and classical classifiers. We use their data, but evaluate on a split grouped by recording so that windows of one recording never sit on both sides of the split.

Corrugation. Rail corrugation is a periodic wear pattern on the rail head whose wavelength is set by the track and vehicle, so its frequency in the measured signal rises with train speed [2]. Soto-Ocampo et al. [3] detect corrugation from on-board acoustic measurements with a mathematical method and publish the recordings with the corrugated sections they confirmed by inspection [4]. We use these recordings and the reported sections as labels.

Rail-VIVID. Ma et al. [5] released synchronised accelerometer, camera and GNSS data from twenty runs over the same 1.4 km of track, with nine surveyed anomalies (rail joints and switch points) given by position. We use only the vibration data.

Pretrained audio models. The Audio Spectrogram Transformer (AST) [6], pretrained on AudioSet [7], and CNNs such as PANNs [8] are strong starting points for small sound-classification datasets. We fine-tune AST.

Multi-task learning. Sharing one encoder across tasks can help or hurt depending on which tasks are trained together [9]. Our shared model is an example of the latter.

III. Data

Table I summarises the three datasets and the four tasks drawn from them.

Rail surface. 720 clips of 17 s (240 per class) recorded at 22.05 kHz and resampled to 16 kHz. Recordings whose file numbers fall in the same block of ten are grouped together, since neighbouring recordings may come from the same stretch of track; groups are split 70/15/15 within each class (481/120/119 clips).

Corrugation. Four runs of an instrumented train over the same 2.4 km at 40, 50, 60 and 75 km/h. The recordings carry kilometre positions but no labels; we label as *corrugated* any clip lying at least 75% inside one of the two sections reported as severely corrugated in [3] (km 4.855–4.914 and 5.239–5.326), and as *normal* any clip lying at least 20 m from both. Clips in between are dropped. The 40 and 50 km/h runs are used for training, 60 km/h for validation and 75 km/h for testing, so every test score is measured at a speed the model never saw. The microphone and axle-box signals are kept as separate tasks. Since the corrugated track is only about 146 m long, corrugated clips overlap (2 m hop); the test run has 62 corrugated and 112 normal clips.

Joints and switches. Rail-VIVID provides 20 runs (five speed and direction settings, four repeats each) of six accelerometers at 2 kHz with GNSS. Section IV-B explains how we label it. We keep the two channels whose vibration repeats between runs and export them as separate clips of the same group. Repeats 1–2 are used for training, 3 for validation and 4 for testing (470 joint and 188 normal test clips).

IV. Method

A. Distance-normalised clips

A corrugation wavelength λ produces a tone at $f = v/\lambda$, so the same defect sounds different at each train speed v. Before cutting clips, we resample every run onto a uniform distance grid using its position record, as if the train had always run at a reference speed v_{ref}. A clip is then a fixed length of track (20 m), and a defect produces the same frequency at any speed. We use $v_{\mathrm{ref}} = 20$ m/s for the UPM data (a 20 m clip lasts 1 s) and 6.5 m/s for Rail-VIVID, whose runs are between 5 and 8.4 m/s. An earlier version that cut clips in time predicted every 75 km/h test clip as normal; with distance normalisation the same models detect it.

B. Labelling joints and switches without labels

The Rail-VIVID vibration files have no per-sample labels; the nine anomalies are given as GNSS positions. We project every sample onto the arc length along a reference trace of the track and compute, per metre and per run, band powers and slow-signal features for each channel. Joints did not appear as isolated impacts on these sensors. What did repeat in every run were stretches of track, typically 15–30 m long, with distinctly higher 500–990 Hz vibration on two channels. Keeping only features whose per-metre profile correlates between runs, with no use of the labels, a step detector finds these stretches. All nine surveyed anomalies lie inside one of them; a few stretches contain no listed anomaly and may be unlisted features of the track [give the final count of stretches and the chance rate from summary.json, key vivid_detector].

We then label a 20 m clip as *joint* when its centre is within 6 m of a surveyed anomaly, and as *normal* when it is at least 25 m from every anomaly and from every detected stretch that holds none (these may be unlisted joints, so they are left out rather than guessed). Stretches where a channel is stuck at a constant value are skipped. This gives 1995 joint and 836 normal clips.

C. Models

Random forest. Each window is described by 137 features: the mean and standard deviation of 20 MFCCs [10] and their deltas, statistics of RMS level, spectral centroid, bandwidth and roll-off, and 48 log-spaced Welch [11] band powers relative to their mean plus the overall level. The band powers keep narrow spectral peaks such as a corrugation tone, which MFCC means tend to smear. A forest of 400 trees with balanced class weights [12] is trained on 1 s windows with 0.5 s hop.

AST. We fine-tune AST pretrained on AudioSet [6], [7]. Its position embeddings are cut to the window length (5 s for rail surface, 1 s for microphone corrugation) rather than padding to 10 s. Each epoch draws eight random windows per recording with class-balanced sampling, plus speed perturbation ($\pm 15\%$), random gain, additive noise at 10–30 dB SNR, and SpecAugment masks [13]. Training uses AdamW with learning rates of 2×10^{-5} for the encoder and 10^{-3} for the head, a cosine schedule with 10% warm-up, label smoothing 0.1, at most 15 epochs, and early stopping after 4 epochs without validation gain. The checkpoint with the best validation macro-F1 is kept.

D. Clip scores and decision threshold

A clip's class probabilities are the mean over its windows. For the two-class tasks, the decision threshold on the fault probability is chosen to maximise macro-F1 on the validation split; with few fault clips, the default 0.5 threshold can miss the fault class entirely. All reported scores are clip-level test macro-F1 with that tuned threshold, or argmax for the three-class rail-surface task.

E. The predictor

The predictor loads one model per task, the AST or forest that scored best, and returns a decision and class probabilities for each task the input sensor supports: microphone clips are scored for rail-surface faults and microphone corrugation, accelerometer clips for axle-box corrugation and joints. Models are trained on separate datasets, so each task only compares a fault with normal track recorded the same way.

V. Results

Every model was trained with different random seeds (three seeds for the forests, two for the AST models) on the same fixed splits. Table II gives the mean and standard deviation of the test macro-F1 and Fig. 1 shows the spread.

TABLE I

DATASETS AND TASKS. EVERY SPLIT IS GROUPED, SO ALL CLIPS FROM ONE RECORDING (OR ONE RUN) STAY IN THE SAME SPLIT. CLIP COUNTS ARE AFTER CUTTING.

Task	Dataset	Sensor	Classes	Clips	Test	Split grouped by
Rail surface	Pakistan Railways [1]	microphone, 16 kHz	normal / superelev. / wheel burn	720	119	recording block
Corrugation (mic)	UPM corrugation [4]	cabin microphone, 16 kHz	normal / corrugated	694	174	run (speed)
Corrugation (axle)	UPM corrugation [4]	axle-box accel., 4 kHz	normal / corrugated	694	174	run (speed)
Joint / switch	Rail-VIVID [5]	accel., 2 kHz	normal / joint	2831	658	repeat

TABLE II

TEST MACRO-F1, MEAN ± STANDARD DEVIATION OVER SEEDS (3 FOR RANDOM FORESTS, 2 FOR AST). THE LAST COLUMN IS THE MODEL THE PREDICTOR USES.

Task	Forest	AST	Predictor uses
Rail surface	0.958 ± 0.009	**0.992 ± 0.012**	AST
Corrugation (axle)	**0.924 ± 0.009**	–	Forest
Corrugation (mic)	0.869 ± 0.028	**0.893 ± 0.038**	AST
Joint / switch	**0.809 ± 0.020**	–	Forest

TABLE III

TEST MACRO-F1 OF THE SHARED MULTI-TASK MODEL IN FOUR RUNS, AGAINST THE SEPARATE MODEL THE PREDICTOR USES (MEAN OVER SEEDS).

Task	Separate	Run 1	Run 2	Run 3	Run 4
Rail surface	**0.99**	0.76	**0.99**	0.56	0.59
Corrugation (axle)	**0.92**	0.83	0.91	0.88	0.86
Corrugation (mic)	**0.89**	0.85	0.75	0.82	0.88
Joint / switch	**0.81**	0.67	0.68	0.66	0.68

figures/scores.pdf
(copy in from paper_results.zip)

Fig. 1. Test macro-F1 per task. Bars are means over seeds; whiskers span the lowest and highest seed.

Rail surface. AST reaches 0.983 and 1.000 on its two seeds, above the forest (0.958), so the predictor uses AST. Window length matters here: in the shared-model runs (Section VI), 1 s windows were too short to tell wheel burns from normal track, whereas the separate AST uses 5 s windows.

Corrugation at an unseen speed. Both sensors detect corrugation on the 75 km/h test run after training on 40 and 50 km/h. The axle-box forest is the most consistent model in the study (0.92 ± 0.01) and finds every corrugated test clip in the last seed (recall 1.00, at 0.88 recall on normal track). For the microphone, AST is ahead of the forest on average but varies more between seeds. In the second seed its validation score peaked after the first epoch and then fell until early stopping (Fig. 3), so it overfits quickly on this small set.

Thresholds matter. With the default 0.5 threshold, the third microphone forest predicted every test clip as normal (macro-F1 0.39); with the threshold tuned on the validation run it scored 0.90. Every result here uses the tuned threshold.

Joints and switches. This is the weakest task. The forest finds nearly all joint clips (recall about 0.96–0.98) but calls a third to half of the normal clips joints (normal recall 0.55–0.66; Fig. 2). AST trained on the same clips did no better in the multi-task runs (Section VI). We think the labels are the limit: a normal clip can contain an unlisted feature of the track, and the 6 m labelling radius around each anomaly is our choice, not a measured extent of the joint.

VI. A SHARED MODEL DID NOT HELP

Our first design was one shared AST encoder with a small output head per task, trained on all four tasks at once. Each head compares a fault only with normal track from its own dataset, so the model cannot score well by recognising which dataset a clip came from. Accelerometer signals were treated as if sampled at 4 kHz so that their content falls within AST's frequency range while a clip stays longer than the window. Each batch picked a task, then a class, then a recording.

Over four runs we changed the window length (1 s to 2 s), the accelerometer speed-up, the training length, gave the microphone corrugation its own head, weighted the rail-surface task twice, and kept the checkpoint whose weakest task scored best on validation rather than the best mean. Only one run matched a separate model on any task (rail surface in run 2), and in that same run microphone corrugation fell to 0.75 (Table III, Fig. 4). The rail-surface task was the most fragile: in runs 3 and 4 the model all but stopped predicting wheel burns, and macro-F1 fell below 0.6. When axle and microphone corrugation shared one head, the microphone score dropped as the axle score rose. The joint task stayed at 0.66–0.68 in every run, below the forest.

With datasets this small and this different, the tasks compete for the shared encoder rather than help each other. Separate models are also simpler to retrain one at a time when new data arrives for one fault, which is why we package them as one predictor instead.

VII. LIMITATIONS

Each dataset covers one line, one vehicle and one sensor setup, so none of these scores shows that a model transfers to other track or other trains. The corrugation labels mark only the two sections reported as severely corrugated; "normal" means not reported as corrugated rather than inspected and

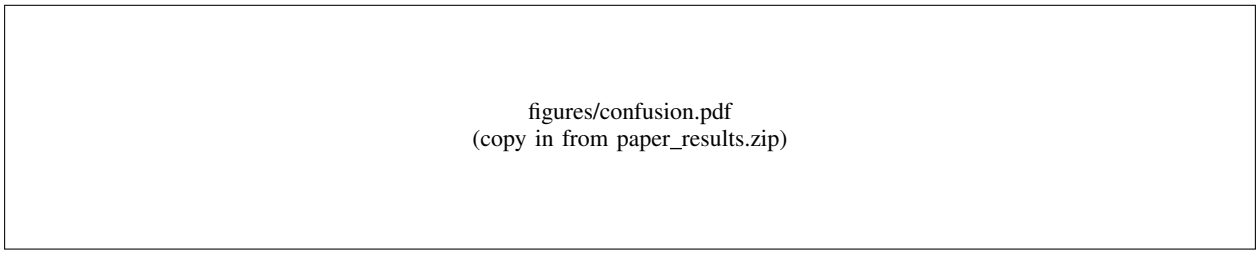


Fig. 2. Test confusion matrices of the model the predictor uses for each task, summed over seeds. Colour is the share of each true class; numbers are clip counts.

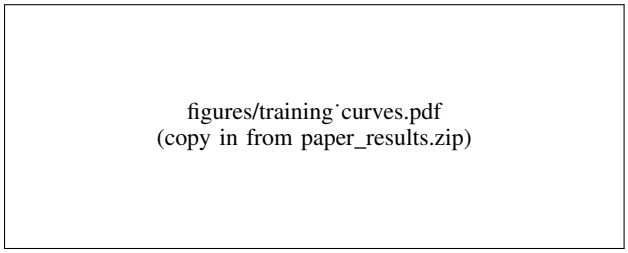


Fig. 3. Validation macro-F1 per epoch of the two AST models, one line per seed.

clean. The joint labels come from surveyed positions and an unsupervised detector, and joint scores are the weakest in the study. The AST models were trained with two seeds rather than three because of limited GPU time. Finally, the four tasks come from different recordings, so the predictor has not been evaluated on a single recording that contains several kinds of fault.

VIII. Conclusion

Public vibration and sound data is enough to build a working predictor for four rail-fault tasks, reaching test macro-F1 of 0.99 on rail-surface faults, 0.92 and 0.89 for corrugation at an unseen speed, and 0.81 on joints. Two simple steps did most of the work: cutting clips by distance rather than time, and tuning the decision threshold on validation data. A shared multi-task model did not match the separate models. The clearest next steps are better joint labels, recordings from more lines and vehicles, and a test on a train carrying both a microphone and accelerometers.

Data and Code Availability

All datasets are public [1], [4], [5]. Code to prepare the clips, train every model, rebuild the tables and figures, and run the predictor is at https://github.com/Zcodeod-OG/Research-Projects (folder `Rail-V/vibration`).

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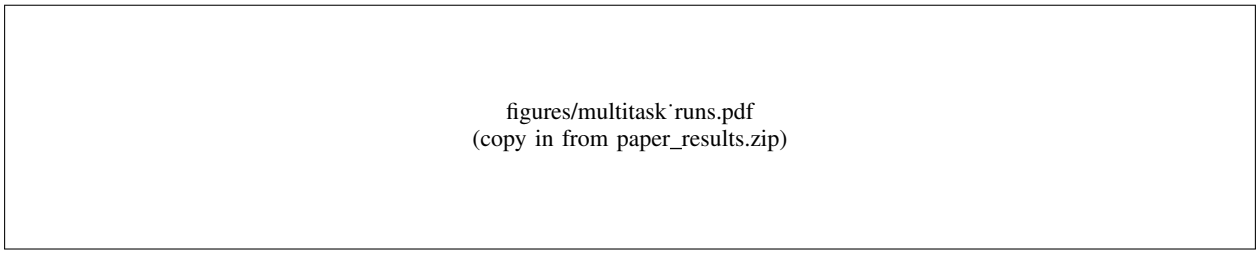


Fig. 4. Test macro-F1 of each shared-model run (solid) against the separate model the predictor uses (dashed).